

A04 — What is Raytracing?

A Short Report on Raytracing: Definition, Applications, and Examples

Student	Team	Course	Due Date
Joseph Clay	TuringCollective	ITAI-1370	Jun 27, 2026

1. What is Raytracing?

Raytracing is a rendering technique that simulates the physical behavior of light by tracing the path of individual light rays as they travel through a virtual three-dimensional scene. Rather than approximating how light behaves — as older rendering methods do — raytracing follows each ray from the camera's viewpoint through every pixel on the screen, tracking what surfaces it strikes, how it reflects, refracts, or is absorbed, and ultimately tracing it back to a light source (NVIDIA, 2024).

The process works in reverse of how real light travels. Instead of simulating every photon emitted from a light source — which would be computationally impossible — raytracing shoots rays outward from the camera through each pixel and calculates what color and illumination that pixel should display based on the objects and light sources the ray encounters. This reverse approach is far more efficient while still producing physically accurate results (Educative.io, 2024).

When a ray hits a surface, several things can happen simultaneously: the ray may **reflect** off a shiny surface (producing accurate mirror-like reflections), **refract** through a transparent material like glass or water (bending the light path), or generate **shadow rays** cast toward light sources to determine whether a point is in shadow. This recursive process repeats for every surface interaction, building up a final pixel color that is physically grounded in real optical behavior.

The fundamental difference between raytracing and the older method of **rasterization** — which dominated computer graphics for decades — is how light is handled. Rasterization uses pre-baked lighting approximations, screen-space tricks, and static shadow maps to simulate light at high speed but with visual compromises. Raytracing eliminates these approximations entirely, producing images that are nearly indistinguishable from photographs — at the cost of significantly higher computational demand (Digital Trends, 2025).

2. Where is Raytracing Used?

Raytracing has existed as a concept since the 1960s, but practical, real-time implementation only became possible in recent years with advances in GPU architecture. Today it spans multiple industries:

- **Video Games:** Real-time raytracing in games became viable in 2018 when NVIDIA launched its RTX platform and Microsoft released the DirectX Raytracing (DXR) framework. These technologies enabled GPUs to calculate raytracing calculations in parallel at speeds fast enough for interactive play. Games now use it for dynamic reflections, accurate ambient occlusion, and global illumination (Unreal Engine, 2024).
 - **Film and Visual Effects:** Hollywood has used raytracing in offline rendering pipelines for decades. CGI in films is rendered frame by frame — sometimes taking hours per frame on render farms — allowing highly complex raytracing calculations that would be impossible in real time. Studios use raytracing to achieve photorealistic lighting in animated and live-action visual effects.
 - **Architecture and Product Visualization:** Architects, interior designers, and product engineers use raytracing to produce photorealistic renders of buildings, rooms, and products before they are built or manufactured. This allows clients to see a finished result with accurate lighting, material textures, and shadows during the design phase.
 - **Virtual and Augmented Reality:** Raytracing enhances immersion in VR and AR by ensuring that virtual objects respond to light in a physically believable way, making digital overlays more convincing when placed in real-world environments.
 - **Scientific and AI Simulation:** Raytracing is used in physics simulations, optics research, and as a tool for generating synthetic training data for computer vision AI systems — producing photorealistic labeled images that would be expensive or impossible to capture in the real world.
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3. Real-World Examples

Cyberpunk 2077 — CD Projekt Red (2020)

One of the most well-known demonstrations of real-time raytracing in gaming. Cyberpunk 2077 uses NVIDIA RTX technology to render dynamic reflections across rain-slicked streets, neon signs, and glass surfaces in real time. The game supports full path tracing (an advanced form of raytracing) through NVIDIA's "Overdrive Mode," producing lighting that rivals pre-rendered CGI (Blender Deluxe, 2025).

Minecraft RTX — Microsoft / Mojang (2020)

Perhaps the most striking visual transformation enabled by raytracing. The blocky, pixelated world of Minecraft — originally built on flat, static lighting — was updated with full raytracing support via NVIDIA RTX. Water became transparent and reflective, torches cast dynamic shadows on surrounding blocks, and sunlight streamed realistically through openings. The result demonstrated that raytracing could transform even stylized games into photorealistic environments (Digital Trends, 2025).

Film CGI — Pixar, ILM, Weta Digital

Feature animation and visual effects studios have relied on raytracing-based renderers for decades. Pixar's RenderMan and Weta Digital's proprietary pipeline use offline raytracing to produce the photorealistic fur, water, fabric, and lighting seen in films like *Avatar*, *The Lion King* (2019), and every Pixar feature. Individual frames can require hours of computation on large render server farms, making quality the priority over speed.

Architectural Visualization — Unreal Engine / V-Ray

Architects and real estate developers use tools like V-Ray and Unreal Engine's Lumen system (which incorporates raytracing) to render photorealistic walkthroughs of buildings that do not yet exist. Clients can experience accurate daylight conditions, shadow patterns throughout the day, and material finishes in a fully interactive virtual environment before a single brick is laid (Unreal Engine, 2024).

Conclusion

Raytracing represents a fundamental shift in how computers generate images — from fast approximations to physically accurate light simulation. What began as an offline tool used only by Hollywood studios over decades of compute time has become a real-time technology available in consumer gaming hardware. Its applications now span entertainment, design, science, and AI, and its influence on how machines represent and understand the visual world continues to grow. As GPU power increases and AI-accelerated denoising techniques (such as NVIDIA's DLSS) reduce its computational cost, raytracing is becoming the default standard for realistic computer-generated imagery across all industries.

Group Members — TuringCollective

The following team members participated in the research and preparation of this assignment:

- Joseph Clay — Primary author, research, and report writing

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